afc2009_mos

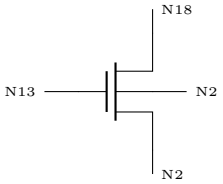
Family `afc2009_mos-21ea6444ebfa` Declaration lines 9–7145

Source `project/afc2009_20261007/mos_r1/afc2009_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `ref` (N3), `fb` (N4), `rst` (N5), `start` (N6), `s0` (N7), `s1` (N8), `s2` (N9), `s3` (N10), `s4` (N11), `done` (N12), `fail` (N13), `hold` (N14), `fine_enable` (N15), `pfd_reset` (N16), `active` (N17)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

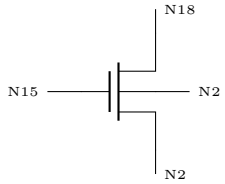
1. `MN3` `n12` source line 10



D N18 = `n1` **G** N13 = `fail` **S** N2 = `vss` **B** N2 = `vss`

`w=0.8u l=0.13u`

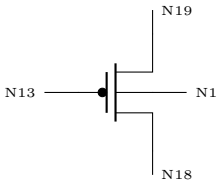
2. `MN4` `n12` source line 11



D N18 = `n1` **G** N15 = `fine_enable` **S** N2 = `vss` **B** N2 = `vss`

`w=0.8u l=0.13u`

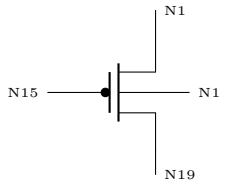
3. `MP5` `p12` source line 12



D N18 = `n1` **G** N13 = `fail` **S** N19 = `n2` **B** N1 = `vdd`

`w=4.8u l=0.13u`

4. `MP6` `p12` source line 13



D N19 = `n2` **G** N15 = `fine_enable` **S** N1 = `vdd` **B** N1 = `vdd`

`w=4.8u l=0.13u`

afc2009_mos

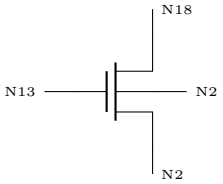
Family `afc2009_mos-a873d37cff17` Declaration lines 8–6917

Source `project/afc2009_20261007/mos/afc2009_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `ref` (N3), `fb` (N4), `rst` (N5), `start` (N6), `s0` (N7), `s1` (N8), `s2` (N9), `s3` (N10), `s4` (N11), `done` (N12), `fail` (N13), `hold` (N14), `fine_enable` (N15), `pfd_reset` (N16), `active` (N17)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

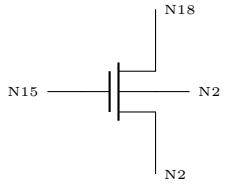
1. `MN3` `n12` source line 9



D N18 = `n1` **G** N13 = `fail` **S** N2 = `vss` **B** N2 = `vss`

`w=0.8u l=0.13u`

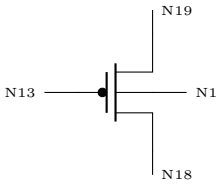
2. `MN4` `n12` source line 10



D N18 = `n1` **G** N15 = `fine_enable` **S** N2 = `vss` **B** N2 = `vss`

`w=0.8u l=0.13u`

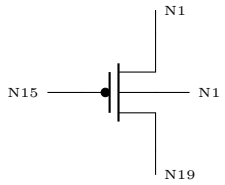
3. `MP5` `p12` source line 11



D N18 = `n1` **G** N13 = `fail` **S** N19 = `n2` **B** N1 = `vdd`

`w=4.8u l=0.13u`

4. `MP6` `p12` source line 12



D N19 = `n2` **G** N15 = `fine_enable` **S** N1 = `vdd` **B** N1 = `vdd`

`w=4.8u l=0.13u`

afc2009_mos

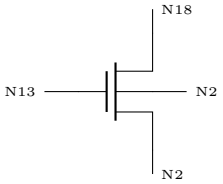
Family `afc2009_mos-fd1ef2a6db68` Declaration lines 10–7850

Source `project/afc2009_20261007/mos_r2/afc2009_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `ref` (N3), `fb` (N4), `rst` (N5), `start` (N6), `s0` (N7), `s1` (N8), `s2` (N9), `s3` (N10), `s4` (N11), `done` (N12), `fail` (N13), `hold` (N14), `fine_enable` (N15), `pfd_reset` (N16), `active` (N17)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

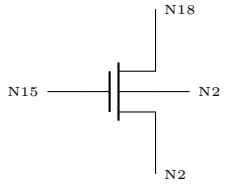
1. `MN3` `n12` source line 11



D N18 = `n1` **G** N13 = `fail` **S** N2 = `vss` **B** N2 = `vss`

`w=0.8u l=0.13u`

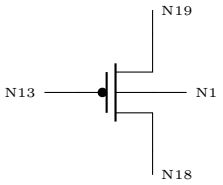
2. `MN4` `n12` source line 12



D N18 = `n1` **G** N15 = `fine_enable` **S** N2 = `vss` **B** N2 = `vss`

`w=0.8u l=0.13u`

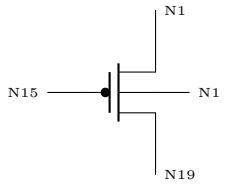
3. `MP5` `p12` source line 13



D N18 = `n1` **G** N13 = `fail` **S** N19 = `n2` **B** N1 = `vdd`

`w=4.8u l=0.13u`

4. `MP6` `p12` source line 14



D N19 = `n2` **G** N15 = `fine_enable` **S** N1 = `vdd` **B** N1 = `vdd`

`w=4.8u l=0.13u`

and2

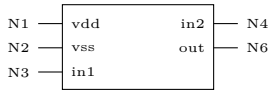
Family `and2-816260a0d071` Declaration lines 44–47

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in1` (N3), `in2` (N4), `out` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XNAND` `nand2` source line 45



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N3 = `in1` **in2** N4 = `in2` **out** N6 = `out_n`

2. `XINV` `inv` source line 46



vdd N1 = `vdd` **vss** N2 = `vss` **in** N6 = `out_n` **out** N5 = `out`

bf_2inv

Family `bf_2inv-97ab11497fc6` Declaration lines 48–51

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in` (N3), `out` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XINV0` `inv` source line 49



vdd N1 = `vdd` **vss** N2 = `vss` **in** N3 = `in` **out** N5 = `n1`

2. `XINV1` `inv` source line 50



vdd N1 = `vdd` **vss** N2 = `vss` **in** N5 = `n1` **out** N4 = `out`

cml_load_receiver

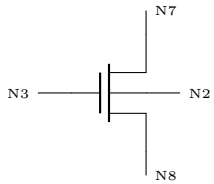
Family `cml_load_receiver-12eed2451cbf` Declaration lines 194–200

Source `project/netlists/receiverload/rx12_c26_v1000000_tran_dad7475efd20f09c_a1.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `inp` (N3), `inn` (N4), `bn` (N5), `op` (N6), `on` (N7)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

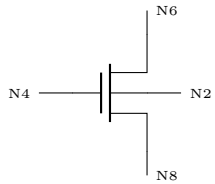
1. `XINP` `n12_ckt_rf` source line 195



D N7 = `on` **G** N3 = `inp` **S** N8 = `rxcs` **B** N2 = `vss`

```
lr=.13u wr=3u nf=4 sar=.4u sbr=.4u mr=1 mismod=0
```

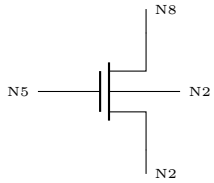
2. `XINN` `n12_ckt_rf` source line 196



D N6 = `op` **G** N4 = `inn` **S** N8 = `rxcs` **B** N2 = `vss`

```
lr=.13u wr=3u nf=4 sar=.4u sbr=.4u mr=1 mismod=0
```

3. `XTAIL` `n12_ckt_rf` source line 197



D N8 = `rxcs` **G** N5 = `bn` **S** N2 = `vss` **B** N2 = `vss`

```
lr=.39u wr=2.5u nf=12 sar=.4u sbr=.4u mr=1 mismod=0
```

4. `RXP` `resistor` source line 198



1 N6 = `op` **2** N1 = `vdd`

```
r=1k
```

cml_load_receiver

Family `cml_load_receiver-12eed2451cbf` Declaration lines 194–200

Source `project/netlists/receiverload/rx12_c26_v1000000_tran_dad7475efd20f09c_a1.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `inp` (N3), `inn` (N4), `bn` (N5), `op` (N6), `on` (N7)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

5. `RXN` `resistor` source line 199



1 N7 = `on` **2** N1 = `vdd`

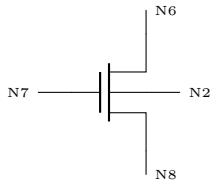
`r=1k`

counter9_gray_cdc_mos

Family `counter9_gray_cdc_mos-7a274891cbd0` Declaration lines 5–2406
Source `project/afc2009_20261007/mos_r2_counter9/counter9_gray_cdc_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `clk` (N3), `ref` (N4), `rst` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

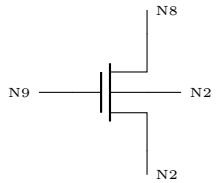
1. `MN3` `n12` source line 6



D N6 = `n1` **G** N7 = `zero_g7` **S** N8 = `n2` **B** N2 = `vss`

`w=1.6u l=0.13u`

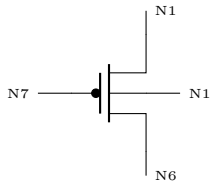
2. `MN4` `n12` source line 7



D N8 = `n2` **G** N9 = `zero_g8` **S** N2 = `vss` **B** N2 = `vss`

`w=1.6u l=0.13u`

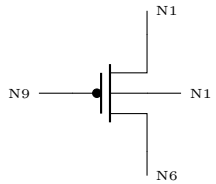
3. `MP5` `p12` source line 8



D N6 = `n1` **G** N7 = `zero_g7` **S** N1 = `vdd` **B** N1 = `vdd`

`w=2.4u l=0.13u`

4. `MP6` `p12` source line 9



D N6 = `n1` **G** N9 = `zero_g8` **S** N1 = `vdd` **B** N1 = `vdd`

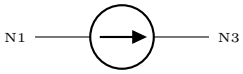
`w=2.4u l=0.13u`

current_ref

Family `current_ref-a648384fc473` Declaration lines 58–63
Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `bp` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

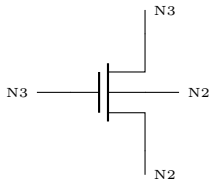
1. `IREF isource` source line 59



1 N1 = `vdd` **2** N3 = `bn`

`dc=IREFVAL`

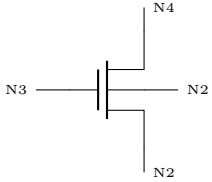
2. `MNREF n12` source line 60



D N3 = `bn` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`w=WREF l=LREF`

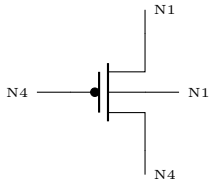
3. `MNPMIRROR n12` source line 61



D N4 = `bp` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`w=WREF l=LREF`

4. `MPREF p12` source line 62



D N4 = `bp` **G** N4 = `bp` **S** N1 = `vdd` **B** N1 = `vdd`

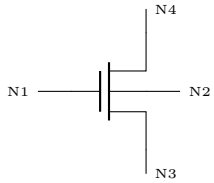
`w=WREFP l=LTAIL`

delay

Family `delay-f95a2e7b256e` Declaration lines 34–37
Source `project/netlists/vco_cmos130_seed.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `in` (N3), `out` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

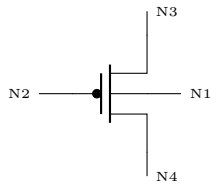
1. `MN n12` source line 35



D N4 = `out` **G** N1 = `vdd` **S** N3 = `in` **B** N2 = `vss`

`w=WLOGICN l=LLOGIC`

2. `MP p12` source line 36



D N4 = `out` **G** N2 = `vss` **S** N3 = `in` **B** N1 = `vdd`

`w=WLOGICP l=LLOGIC`

inv

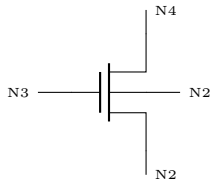
Family `inv-bbc9ea216ec3` Declaration lines 20–23

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in` (N3), `out` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

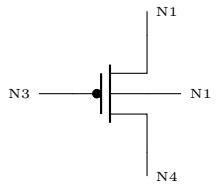
1. `MN n12` source line 21



D N4 = `out` **G** N3 = `in` **S** N2 = `vss` **B** N2 = `vss`

`w=WLOGICN l=LLOGIC`

2. `MP p12` source line 22



D N4 = `out` **G** N3 = `in` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WLOGICP l=LLOGIC`

nand2

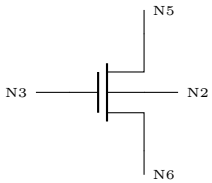
Family `nand2-e08d9ff0185b` Declaration lines 34–39

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in1` (N3), `in2` (N4), `out` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

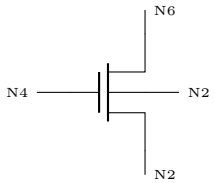
1. `MN1` `n12` source line 35



D N5 = `out` **G** N3 = `in1` **S** N6 = `nseries` **B** N2 = `vss`

`w=WLOGICN l=LLOGIC`

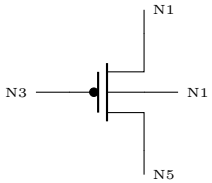
2. `MN2` `n12` source line 36



D N6 = `nseries` **G** N4 = `in2` **S** N2 = `vss` **B** N2 = `vss`

`w=WLOGICN l=LLOGIC`

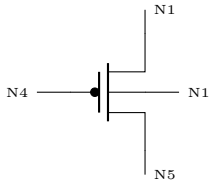
3. `MP1` `p12` source line 37



D N5 = `out` **G** N3 = `in1` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WLOGICP l=LLOGIC`

4. `MP2` `p12` source line 38



D N5 = `out` **G** N4 = `in2` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WLOGICP l=LLOGIC`

nor2

Cards 1–4 of 4

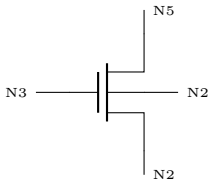
Family `nor2-e9c09461b380` Declaration lines 28–33

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in1` (N3), `in2` (N4), `out` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

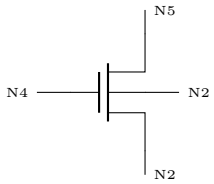
1. `MN1 n12` source line 29



D N5 = `out` **G** N3 = `in1` **S** N2 = `vss` **B** N2 = `vss`

`w=WLOGICN l=LLOGIC`

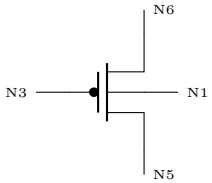
2. `MN2 n12` source line 30



D N5 = `out` **G** N4 = `in2` **S** N2 = `vss` **B** N2 = `vss`

`w=WLOGICN l=LLOGIC`

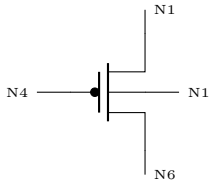
3. `MP1 p12` source line 31



D N5 = `out` **G** N3 = `in1` **S** N6 = `pseries` **B** N1 = `vdd`

`w=WLOGICP l=LLOGIC`

4. `MP2 p12` source line 32



D N6 = `pseries` **G** N4 = `in2` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WLOGICP l=LLOGIC`

or2

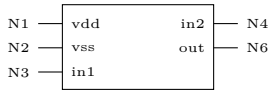
Family `or2-7ab8b69b70dc` Declaration lines 40–43

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in1` (N3), `in2` (N4), `out` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XNOR` `nor2` source line 41



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N3 = `in1` **in2** N4 = `in2` **out** N6 = `out_n`

2. `XINV` `inv` source line 42



vdd N1 = `vdd` **vss** N2 = `vss` **in** N6 = `out_n` **out** N5 = `out`

pll130_and2

Family `pll130_and2-37d0229f5810` Declaration lines 30–33
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `b` (N4), `y` (N5)

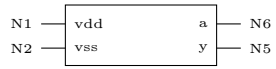
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XN pll130_nand2` source line 31



vdd N1 = `vdd` **vss** N2 = `vss` **a** N3 = `a` **b** N4 = `b` **y** N6 = `yn`

2. `XI pll130_inv` source line 32



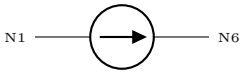
vdd N1 = `vdd` **vss** N2 = `vss` **a** N6 = `yn` **y** N5 = `y`

pll130_cp_drain

Family `pll130_cp_drain-156bce7f8c45` Declaration lines 12–25
Source `project/pll_full/analog/pll130_cp_drain_candidate.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `up` (N3), `dn` (N4), `ctrl` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

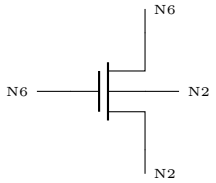
1. `IREF isource` source line 15



1 N1 = `vdd` **2** N6 = `bncp`

`dc=IREF_CP`

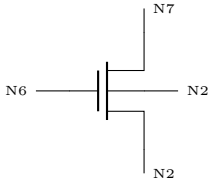
2. `MNREF n12` source line 16



D N6 = `bncp` **G** N6 = `bncp` **S** N2 = `vss` **B** N2 = `vss`

`w=WN_CP l=L_CP`

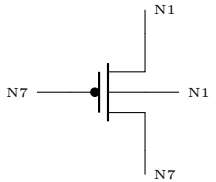
3. `MNPMIRROR n12` source line 17



D N7 = `bpcp` **G** N6 = `bncp` **S** N2 = `vss` **B** N2 = `vss`

`w=WN_CP l=L_CP`

4. `MPREF p12` source line 18



D N7 = `bpcp` **G** N7 = `bpcp` **S** N1 = `vdd` **B** N1 = `vdd`

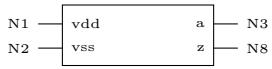
`w=WP_CP l=L_CP`

pll130_cp_drain

Family `pll130_cp_drain-156bce7f8c45` Declaration lines 12–25
Source `project/pll_full/analog/pll130_cp_drain_candidate.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `up` (N3), `dn` (N4), `ctrl` (N5)

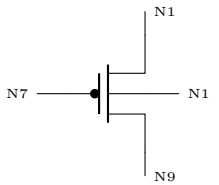
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

5. `XUI` `pll130_cpd_inv` source line 19



vdd N1 = `vdd` **vss** N2 = `vss` **a** N3 = `up` **z** N8 = `upbar`

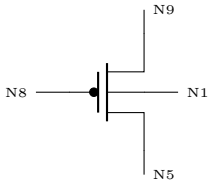
6. `MPUP` `p12` source line 21



D N9 = `pinternal` **G** N7 = `bpcp` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WP_CP l=L_CP`

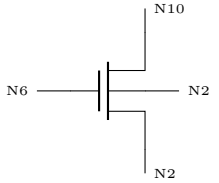
7. `MPUSW` `p12` source line 22



D N5 = `ctrl` **G** N8 = `upbar` **S** N9 = `pinternal` **B** N1 = `vdd`

`w=WP_SW l=L_SW`

8. `MNDN` `n12` source line 23



D N10 = `ninternal` **G** N6 = `bncp` **S** N2 = `vss` **B** N2 = `vss`

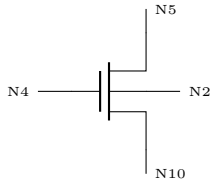
`w=WN_CP l=L_CP`

pll130_cp_drain

Family `pll130_cp_drain-156bce7f8c45` Declaration lines 12–25
Source `project/pll_full/analog/pll130_cp_drain_candidate.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `up` (N3), `dn` (N4), `ctrl` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

9. `MNDSW n12` source line 24



D N5 = `ctrl` **G** N4 = `dn` **S** N10 = `ninternal` **B** N2 = `vss`

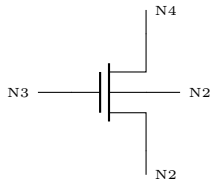
`w=WN_SW l=L_SW`

pll130_cp_inv

Family `pll130_cp_inv-28e90483c781` Declaration lines 7–10
Source `project/pll_full/analog/pll130_cp_transistor.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `z` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

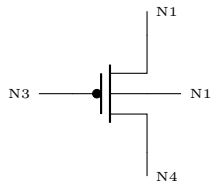
1. `MN n12` source line 8



D N4 = `z` **G** N3 = `a` **S** N2 = `vss` **B** N2 = `vss`

`w=0.5u l=0.13u`

2. `MP p12` source line 9



D N4 = `z` **G** N3 = `a` **S** N1 = `vdd` **B** N1 = `vdd`

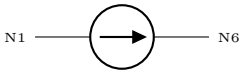
`w=1.5u l=0.13u`

pll130_cp_transistor

Family `pll130_cp_transistor-8894127cf4d2` Declaration lines 12–36
Source `project/pll_full/analog/pll130_cp_transistor.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `up` (N3), `dn` (N4), `ctrl` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

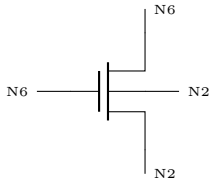
1. `IREF isource` source line 15



1 N1 = `vdd` **2** N6 = `bncp`

```
dc=IREF_CP
```

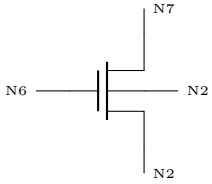
2. `MNREF n12` source line 16



D N6 = `bncp` **G** N6 = `bncp` **S** N2 = `vss` **B** N2 = `vss`

```
w=WN_CP l=L_CP
```

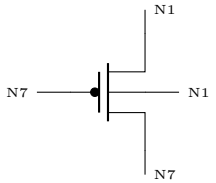
3. `MNPMIRROR n12` source line 17



D N7 = `bpcp` **G** N6 = `bncp` **S** N2 = `vss` **B** N2 = `vss`

```
w=WN_CP l=L_CP
```

4. `MPREF p12` source line 18



D N7 = `bpcp` **G** N7 = `bpcp` **S** N1 = `vdd` **B** N1 = `vdd`

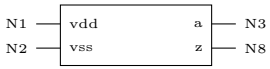
```
w=WP_CP l=L_CP
```

pll130_cp_transistor

Family `pll130_cp_transistor-8894127cf4d2` Declaration lines 12–36
Source `project/pll_full/analog/pll130_cp_transistor.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `up` (N3), `dn` (N4), `ctrl` (N5)

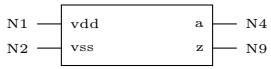
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

5. `XUPINV` `pll130_cp_inv` source line 20



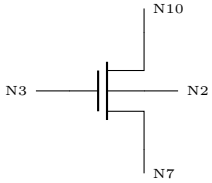
vdd N1 = `vdd` **vss** N2 = `vss` **a** N3 = `up` **z** N8 = `upbar`

6. `XDNINV` `pll130_cp_inv` source line 21



vdd N1 = `vdd` **vss** N2 = `vss` **a** N4 = `dn` **z** N9 = `dnbar`

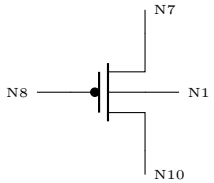
7. `MNUPBIAS` `n12` source line 24



D N10 = `gp` **G** N3 = `up` **S** N7 = `bpcp` **B** N2 = `vss`

`w=WN_SW l=L_SW`

8. `MPUPBIAS` `p12` source line 25



D N10 = `gp` **G** N8 = `upbar` **S** N7 = `bpcp` **B** N1 = `vdd`

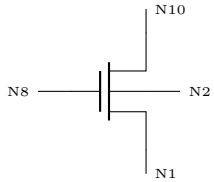
`w=WP_SW l=L_SW`

pll130_cp_transistor

Family `pll130_cp_transistor-8894127cf4d2` Declaration lines 12–36
Source `project/pll_full/analog/pll130_cp_transistor.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `up` (N3), `dn` (N4), `ctrl` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

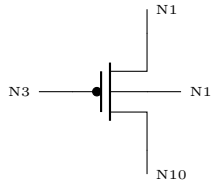
9. `MNUPRAIL n12` source line 26



D N10 = `gp` **G** N8 = `upbar` **S** N1 = `vdd` **B** N2 = `vss`

```
w=WN_SW l=L_SW
```

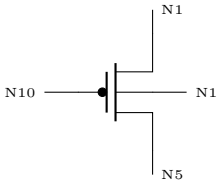
10. `MPUPRAIL p12` source line 27



D N10 = `gp` **G** N3 = `up` **S** N1 = `vdd` **B** N1 = `vdd`

```
w=WP_SW l=L_SW
```

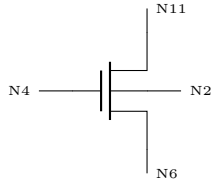
11. `MPUP p12` source line 28



D N5 = `ctrl` **G** N10 = `gp` **S** N1 = `vdd` **B** N1 = `vdd`

```
w=WP_CP l=L_CP
```

12. `MNDNBIAS n12` source line 31



D N11 = `gn` **G** N4 = `dn` **S** N6 = `bncp` **B** N2 = `vss`

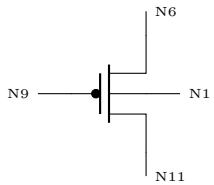
```
w=WN_SW l=L_SW
```

pll130_cp_transistor

Family `pll130_cp_transistor-8894127cf4d2` Declaration lines 12–36
Source `project/pll_full/analog/pll130_cp_transistor.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `up` (N3), `dn` (N4), `ctrl` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

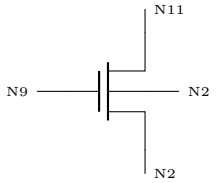
13. `MPDNBIAS` `p12` source line 32



D N11 = `gn` **G** N9 = `dnbar` **S** N6 = `bncp` **B** N1 = `vdd`

```
w=WP_SW l=L_SW
```

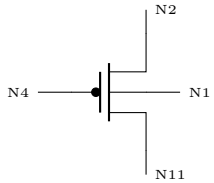
14. `MNDNRAIL` `n12` source line 33



D N11 = `gn` **G** N9 = `dnbar` **S** N2 = `vss` **B** N2 = `vss`

```
w=WN_SW l=L_SW
```

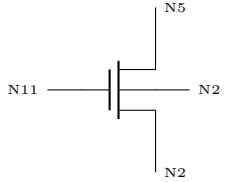
15. `MPDNRAIL` `p12` source line 34



D N11 = `gn` **G** N4 = `dn` **S** N2 = `vss` **B** N1 = `vdd`

```
w=WP_SW l=L_SW
```

16. `MNDN` `n12` source line 35



D N5 = `ctrl` **G** N11 = `gn` **S** N2 = `vss` **B** N2 = `vss`

```
w=WN_CP l=L_CP
```

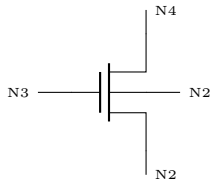
pll130_cpd_inv

Cards 1–2 of 2

Family `pll130_cpd_inv-d97eee629c12` Declaration lines 7–10
Source `project/pll_full/analog/pll130_cp_drain_candidate.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `z` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

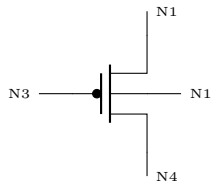
1. `MN n12` source line 8



D N4 = `z` **G** N3 = `a` **S** N2 = `vss` **B** N2 = `vss`

`w=0.8u l=0.13u`

2. `MP p12` source line 9



D N4 = `z` **G** N3 = `a` **S** N1 = `vdd` **B** N1 = `vdd`

`w=2.4u l=0.13u`

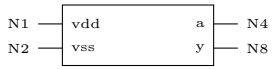
pll130_dff

Cards 1–4 of 8

Family `pll130_dff-ffbf8308013f` Declaration lines 82–91
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `d` (N3), `clk` (N4), `rst` (N5), `q` (N6), `qb` (N7)

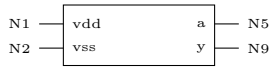
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XCLK` `pll130_inv` source line 83



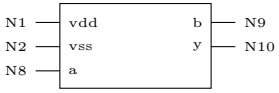
vdd N1 = `vdd` **vss** N2 = `vss` **a** N4 = `clk` **y** N8 = `clkb`

2. `XRST` `pll130_inv` source line 84



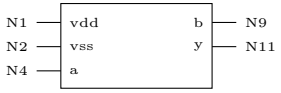
vdd N1 = `vdd` **vss** N2 = `vss` **a** N5 = `rst` **y** N9 = `rstb`

3. `XL` `pll130_and2` source line 85



vdd N1 = `vdd` **vss** N2 = `vss` **a** N8 = `clkb` **b** N9 = `rstb` **y** N10 = `low_en`

4. `XH` `pll130_and2` source line 86



vdd N1 = `vdd` **vss** N2 = `vss` **a** N4 = `clk` **b** N9 = `rstb` **y** N11 = `high_en`

pll130_dff

Family `pll130_dff-ffbf8308013f` Declaration lines 82–91
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `d` (N3), `clk` (N4), `rst` (N5), `q` (N6), `qb` (N7)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

5. `XLN` `pll130_inv` source line 87



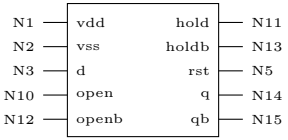
vdd N1 = `vdd` **vss** N2 = `vss` **a** N10 = `low_en` **y** N12 = `low_enb`

6. `XHN` `pll130_inv` source line 88



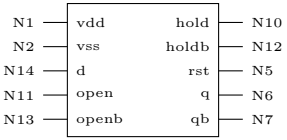
vdd N1 = `vdd` **vss** N2 = `vss` **a** N11 = `high_en` **y** N13 = `high_enb`

7. `XM` `pll130_latch` source line 89



vdd N1 = `vdd` **vss** N2 = `vss` **d** N3 = `d` **open** N10 = `low_en` **openb** N12 = `low_enb`
hold N11 = `high_en` **holdb** N13 = `high_enb` **rst** N5 = `rst` **q** N14 = `qm` **qb** N15 = `qmb`

8. `XS` `pll130_latch` source line 90



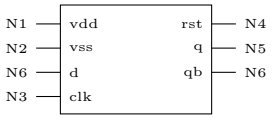
vdd N1 = `vdd` **vss** N2 = `vss` **d** N14 = `qm` **open** N11 = `high_en` **openb** N13 = `high_enb`
hold N10 = `low_en` **holdb** N12 = `low_enb` **rst** N5 = `rst` **q** N6 = `q` **qb** N7 = `qb`

pll130_div2

Family `pll130_div2-a78c881ddf29` Declaration lines 93–95
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `clk` (N3), `rst` (N4), `q` (N5), `qb` (N6)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XF` `pll130_dff` source line 94



vdd N1 = `vdd` **vss** N2 = `vss` **d** N6 = `qb` **clk** N3 = `clk` **rst** N4 = `rst` **q** N5 = `q`
qb N6 = `qb`

pll130_div32_cmos

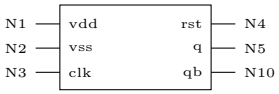
Family `pll130_div32_cmos-22d9bd5f3682` Declaration lines 99–105

Source `project/pll_full/digital/pll130_digital_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `clk` (N3), `rst` (N4), `q1` (N5), `q2` (N6), `q3` (N7), `q4` (N8), `q5` (N9)

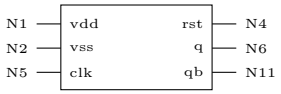
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XD0` `pll130_div2` source line 100



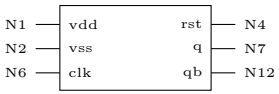
vdd N1 = `vdd` **vss** N2 = `vss` **clk** N3 = `clk` **rst** N4 = `rst` **q** N5 = `q1` **qb** N10 = `q1b`

2. `XD1` `pll130_div2` source line 101



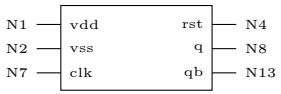
vdd N1 = `vdd` **vss** N2 = `vss` **clk** N5 = `q1` **rst** N4 = `rst` **q** N6 = `q2` **qb** N11 = `q2b`

3. `XD2` `pll130_div2` source line 102



vdd N1 = `vdd` **vss** N2 = `vss` **clk** N6 = `q2` **rst** N4 = `rst` **q** N7 = `q3` **qb** N12 = `q3b`

4. `XD3` `pll130_div2` source line 103



vdd N1 = `vdd` **vss** N2 = `vss` **clk** N7 = `q3` **rst** N4 = `rst` **q** N8 = `q4` **qb** N13 = `q4b`

pll130_div32_cmos

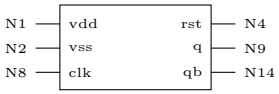
Family `pll130_div32_cmos-22d9bd5f3682` Declaration lines 99–105

Source `project/pll_full/digital/pll130_digital_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `clk` (N3), `rst` (N4), `q1` (N5), `q2` (N6), `q3` (N7), `q4` (N8), `q5` (N9)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

5. `XD4` `pll130_div2` source line 104



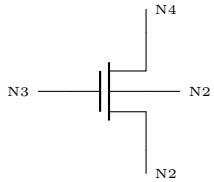
vdd N1 = `vdd` **vss** N2 = `vss` **clk** N8 = `q4` **rst** N4 = `rst` **q** N9 = `q5` **qb** N14 = `q5b`

pll130_inv

Family `pll130_inv-6b241e6f2334` Declaration lines 8–12
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `y` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

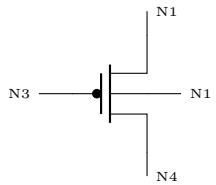
1. `MN n12` source line 10



D N4 = `y` **G** N3 = `a` **S** N2 = `vss` **B** N2 = `vss`

`w=WN l=LL`

2. `MP p12` source line 11



D N4 = `y` **G** N3 = `a` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WP l=LL`

pll130_latch

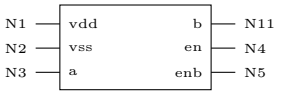
Family `pll130_latch-4356a8a9d324` Declaration lines 71–77

Source `project/pll_full/digital/pll130_digital_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `d` (N3), `open` (N4), `openb` (N5), `hold` (N6), `holdb` (N7), `rst` (N8), `q` (N9), `qb` (N10)

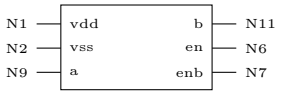
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XIN` `pll130_tg` source line 72



vdd N1 = `vdd` **vss** N2 = `vss` **a** N3 = `d` **b** N11 = `storage` **en** N4 = `open`
enb N5 = `openb`

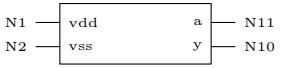
2. `XFB` `pll130_tg` source line 73



vdd N1 = `vdd` **vss** N2 = `vss` **a** N9 = `q` **b** N11 = `storage` **en** N6 = `hold`
enb N7 = `holdb`

---

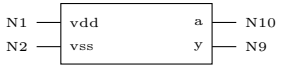
3. `XINV0` `pll130_inv` source line 74



vdd N1 = `vdd` **vss** N2 = `vss` **a** N11 = `storage` **y** N10 = `qb`

`WN=.5u WP=1.5u`

4. `XINV1` `pll130_inv` source line 75



vdd N1 = `vdd` **vss** N2 = `vss` **a** N10 = `qb` **y** N9 = `q`

`WN=.5u WP=1.5u`

pll130_latch

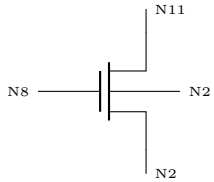
Family `pll130_latch-4356a8a9d324` Declaration lines 71–77

Source `project/pll_full/digital/pll130_digital_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `d` (N3), `open` (N4), `openb` (N5), `hold` (N6), `holdb` (N7), `rst` (N8), `q` (N9), `qb` (N10)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

5. `MRST n12` source line 76



D N11 = `storage` **G** N8 = `rst` **S** N2 = `vss` **B** N2 = `vss`

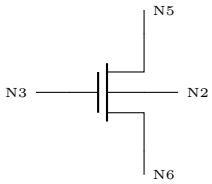
`w=2u l=.13u`

pll130_nand2

Family `pll130_nand2-00971f9d0809` Declaration lines 14–20
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `b` (N4), `y` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

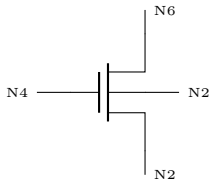
1. `MN0 n12` source line 16



D N5 = `y` **G** N3 = `a` **S** N6 = `ns` **B** N2 = `vss`

`w=WN l=LL`

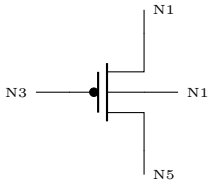
2. `MN1 n12` source line 17



D N6 = `ns` **G** N4 = `b` **S** N2 = `vss` **B** N2 = `vss`

`w=WN l=LL`

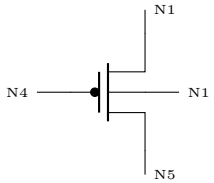
3. `MP0 p12` source line 18



D N5 = `y` **G** N3 = `a` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WP l=LL`

4. `MP1 p12` source line 19



D N5 = `y` **G** N4 = `b` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WP l=LL`

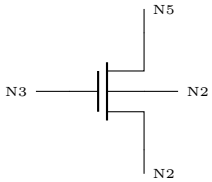
pll130_nor2

Cards 1–4 of 4

Family `pll130_nor2-c366bab03699` Declaration lines 22–28
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `b` (N4), `y` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

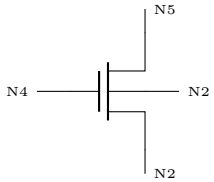
1. `MN0 n12` source line 24



D N5 = `y` **G** N3 = `a` **S** N2 = `vss` **B** N2 = `vss`

`w=WN l=LL`

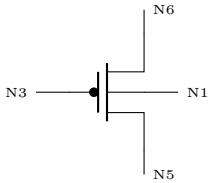
2. `MN1 n12` source line 25



D N5 = `y` **G** N4 = `b` **S** N2 = `vss` **B** N2 = `vss`

`w=WN l=LL`

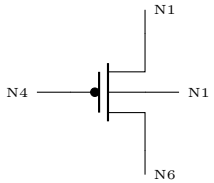
3. `MP0 p12` source line 26



D N5 = `y` **G** N3 = `a` **S** N6 = `ps` **B** N1 = `vdd`

`w=WP l=LL`

4. `MP1 p12` source line 27



D N6 = `ps` **G** N4 = `b` **S** N1 = `vdd` **B** N1 = `vdd`

`w=WP l=LL`

pll130_or2

Family `pll130_or2-d0f56e0a1570` Declaration lines 35–38
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `b` (N4), `y` (N5)

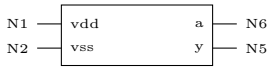
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XN pll130_nor2` source line 36



vdd N1 = `vdd` **vss** N2 = `vss` **a** N3 = `a` **b** N4 = `b` **y** N6 = `yn`

2. `XI pll130_inv` source line 37



vdd N1 = `vdd` **vss** N2 = `vss` **a** N6 = `yn` **y** N5 = `y`

pll130_pfd

Cards 1–4 of 8

Family `pll130_pfd-bd7572bf5d53` Declaration lines 52–61
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `ref` (N3), `fb` (N4), `rst` (N5), `up` (N6), `dn` (N7)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XU` `pll130_pfd_tspc` source line 53



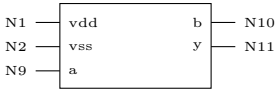
vdd N1 = `vdd` **vss** N2 = `vss` **a** N3 = `ref` **b** N8 = `reset` **y** N9 = `upn`

2. `XD` `pll130_pfd_tspc` source line 54



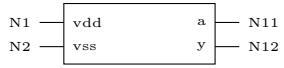
vdd N1 = `vdd` **vss** N2 = `vss` **a** N4 = `fb` **b** N8 = `reset` **y** N10 = `dnn`

3. `XN` `pll130_nor2` source line 55



vdd N1 = `vdd` **vss** N2 = `vss` **a** N9 = `upn` **b** N10 = `dnn` **y** N11 = `rst0`

4. `XI0` `pll130_inv` source line 56



vdd N1 = `vdd` **vss** N2 = `vss` **a** N11 = `rst0` **y** N12 = `rst1`

pll130_pfd

Cards 5–8 of 8

Family `pll130_pfd-bd7572bf5d53` Declaration lines 52–61
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `ref` (N3), `fb` (N4), `rst` (N5), `up` (N6), `dn` (N7)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

5. `XI1` `pll130_inv` source line 57



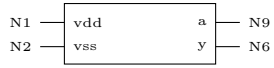
vdd N1 = `vdd` **vss** N2 = `vss` **a** N12 = `rst1` **y** N13 = `rst2`

6. `XRST` `pll130_or2` source line 58



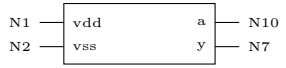
vdd N1 = `vdd` **vss** N2 = `vss` **a** N5 = `rst` **b** N13 = `rst2` **y** N8 = `reset`

7. `XIU` `pll130_inv` source line 59



vdd N1 = `vdd` **vss** N2 = `vss` **a** N9 = `upn` **y** N6 = `up`

8. `XID` `pll130_inv` source line 60



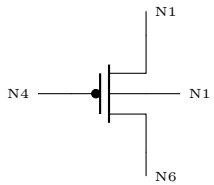
vdd N1 = `vdd` **vss** N2 = `vss` **a** N10 = `dnn` **y** N7 = `dn`

pll130_pfd_tspc

Family `pll130_pfd_tspc-ead1e01bf05c` Declaration lines 43–50
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `b` (N4), `y` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

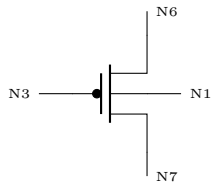
1. `MP0` `p12` source line 44



D N6 = `n1` **G** N4 = `b` **S** N1 = `vdd` **B** N1 = `vdd`

`w=2.4u l=.13u`

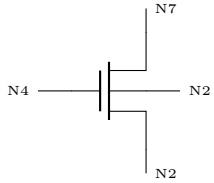
2. `MP1` `p12` source line 45



D N7 = `n2` **G** N3 = `a` **S** N6 = `n1` **B** N1 = `vdd`

`w=2.4u l=.13u`

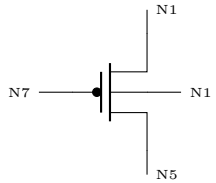
3. `MN0` `n12` source line 46



D N7 = `n2` **G** N4 = `b` **S** N2 = `vss` **B** N2 = `vss`

`w=.8u l=.13u`

4. `MP2` `p12` source line 47



D N5 = `y` **G** N7 = `n2` **S** N1 = `vdd` **B** N1 = `vdd`

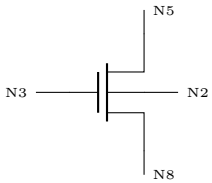
`w=2.4u l=.13u`

pll130_pfd_tspc

Family `pll130_pfd_tspc-ead1e01bf05c` Declaration lines 43–50
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `b` (N4), `y` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

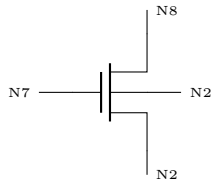
5. `MN1 n12` source line 48



D N5 = `y` **G** N3 = `a` **S** N8 = `n3` **B** N2 = `vss`

`w=.8u l=.13u`

6. `MN2 n12` source line 49



D N8 = `n3` **G** N7 = `n2` **S** N2 = `vss` **B** N2 = `vss`

`w=2.4u l=.13u`

pll130_receiver

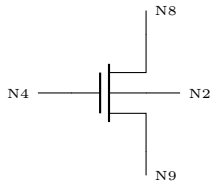
Family `pll130_receiver-ba0a82bb08ea` Declaration lines 112–121

Source `project/pll_full/digital/pll130_digital_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `inp` (N4), `inn` (N5), `clk` (N6), `rxp` (N7), `rxn` (N8)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

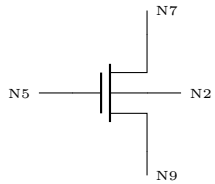
1. `XIP` `n12_ckt_rf` source line 114



D N8 = `rxn` **G** N4 = `inp` **S** N9 = `src` **B** N2 = `vss`

```
lr=.13u wr=INPUT_WR nf=INPUT_NF sar=.4u sbr=.4u mr=1 mismod=0
```

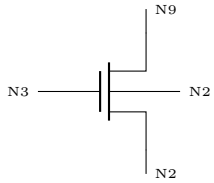
2. `XIN` `n12_ckt_rf` source line 115



D N7 = `rxp` **G** N5 = `inn` **S** N9 = `src` **B** N2 = `vss`

```
lr=.13u wr=INPUT_WR nf=INPUT_NF sar=.4u sbr=.4u mr=1 mismod=0
```

3. `XT` `n12_ckt_rf` source line 116



D N9 = `src` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

```
lr=.39u wr=TAIL_WR nf=TAIL_NF sar=.4u sbr=.4u mr=1 mismod=0
```

4. `RP` `resistor` source line 117



1 N7 = `rxp` **2** N1 = `vdd`

```
r=RLOAD_RX
```

pll130_receiver

Family `pll130_receiver-ba0a82bb08ea` Declaration lines 112–121
Source `project/pll_full/digital/pll130_digital_mos.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `inp` (N4), `inn` (N5), `clk` (N6), `rxp` (N7), `rxn` (N8)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

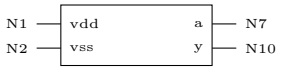
5. `RN` `resistor` source line 118



1 N8 = `rxn` **2** N1 = `vdd`

`r=RLOAD_RX`

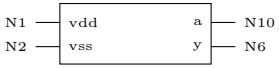
6. `XI0` `pll130_inv` source line 119



vdd N1 = `vdd` **vss** N2 = `vss` **a** N7 = `rxp` **y** N10 = `clkn`

`WN=INVO_WN WP=INVO_WP`

7. `XI1` `pll130_inv` source line 120



vdd N1 = `vdd` **vss** N2 = `vss` **a** N10 = `clkn` **y** N6 = `clk`

`WN=1.6u WP=4.8u`

pll130_receiver_ac

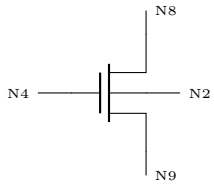
Family `pll130_receiver_ac-e4c86aa67319` Declaration lines 7–18

Source `project/pll_full/digital/pll130_receiver_ac.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `inp` (N4), `inn` (N5), `clk` (N6), `rxp` (N7), `rxn` (N8)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

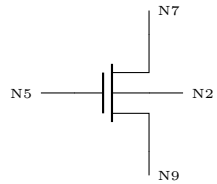
1. `XIP` `n12_ckt_rf` source line 9



D N8 = `rxn` **G** N4 = `inp` **S** N9 = `src` **B** N2 = `vss`

```
lr=.13u wr=INPUT_WR nf=INPUT_NF sar=.4u sbr=.4u mr=1 mismod=0
```

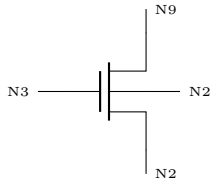
2. `XIN` `n12_ckt_rf` source line 10



D N7 = `rxp` **G** N5 = `inn` **S** N9 = `src` **B** N2 = `vss`

```
lr=.13u wr=INPUT_WR nf=INPUT_NF sar=.4u sbr=.4u mr=1 mismod=0
```

3. `XT` `n12_ckt_rf` source line 11



D N9 = `src` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

```
lr=.39u wr=TAIL_WR nf=TAIL_NF sar=.4u sbr=.4u mr=1 mismod=0
```

4. `RP` `resistor` source line 12



1 N7 = `rxp` **2** N1 = `vdd`

```
r=RLOAD_RX
```

pll130_receiver_ac

Family `pll130_receiver_ac-e4c86aa67319` Declaration lines 7–18

Source `project/pll_full/digital/pll130_receiver_ac.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `inp` (N4), `inn` (N5), `clk` (N6), `rxp` (N7), `rxn` (N8)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

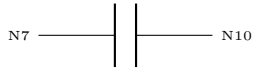
5. `RN` `resistor` source line 13



1 N8 = `rxn` **2** N1 = `vdd`

`r=RLOAD_RX`

6. `CAC` `capacitor` source line 14



1 N7 = `rxp` **2** N10 = `inv_bias`

`c=CC`

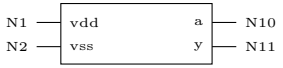
7. `RBIAS` `resistor` source line 15



1 N11 = `clkn` **2** N10 = `inv_bias`

`r=RB`

8. `XI0` `pll130_inv` source line 16



vdd N1 = `vdd` **vss** N2 = `vss` **a** N10 = `inv_bias` **y** N11 = `clkn`

`WN=.8u WP=3.2u`

pll130_receiver_ac

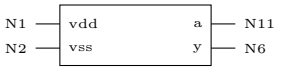
Family `pll130_receiver_ac-e4c86aa67319` Declaration lines 7–18

Source `project/pll_full/digital/pll130_receiver_ac.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `inp` (N4), `inn` (N5), `clk` (N6), `rxp` (N7), `rxn` (N8)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

9. `XI1` `pll130_inv` source line 17



vdd N1 = `vdd` **vss** N2 = `vss` **a** N11 = `clkn` **y** N6 = `clk`

`WN=1.6u WP=4.8u`

pll130_tg

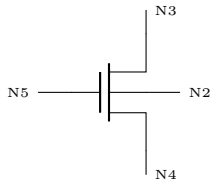
Family `pll130_tg-20004b8c9f9d` Declaration lines 63–66

Source `project/pll_full/digital/pll130_digital_mos.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `a` (N3), `b` (N4), `en` (N5), `enb` (N6)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

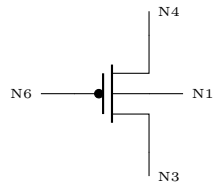
1. `MN n12` source line 64



D N3 = `a` **G** N5 = `en` **S** N4 = `b` **B** N2 = `vss`

```
w=1.2u l=.13u
```

2. `MP p12` source line 65



D N3 = `a` **G** N6 = `enb` **S** N4 = `b` **B** N1 = `vdd`

```
w=3.6u l=.13u
```

s_to_d

Cards 1–4 of 4

Family `s_to_d-39789199da9a` Declaration lines 68–73
Source `project/netlists/vco_cmos130_seed.scs`
Ports in declaration order `vdd` (N1), `vss` (N2), `in` (N3), `out_p` (N4), `out_n` (N5)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XINV0` `inv` source line 69



vdd N1 = `vdd` **vss** N2 = `vss` **in** N3 = `in` **out** N6 = `n1`

2. `XDELAY` `delay` source line 70



vdd N1 = `vdd` **vss** N2 = `vss` **in** N3 = `in` **out** N7 = `n2`

3. `XINV1` `inv` source line 71



vdd N1 = `vdd` **vss** N2 = `vss` **in** N6 = `n1` **out** N4 = `out_p`

4. `XINV2` `inv` source line 72



vdd N1 = `vdd` **vss** N2 = `vss` **in** N7 = `n2` **out** N5 = `out_n`

s_to_d

Family `s_to_d-9232d664ce57` Declaration lines 52–57

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in` (N3), `out_p` (N4), `out_n` (N5)

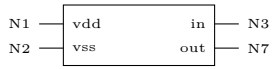
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XINV0` `inv` source line 53



vdd N1 = `vdd` **vss** N2 = `vss` **in** N3 = `in` **out** N6 = `n1`

2. `XDELAY` `tg_delay` source line 54



vdd N1 = `vdd` **vss** N2 = `vss` **in** N3 = `in` **out** N7 = `n2`

3. `XINV1` `inv` source line 55



vdd N1 = `vdd` **vss** N2 = `vss` **in** N6 = `n1` **out** N4 = `out_p`

4. `XINV2` `inv` source line 56



vdd N1 = `vdd` **vss** N2 = `vss` **in** N7 = `n2` **out** N5 = `out_n`

tg_delay

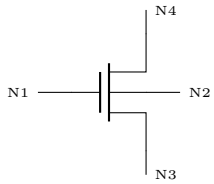
Family `tg_delay-fb6f32dc3b7a` Declaration lines 24–27

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `in` (N3), `out` (N4)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

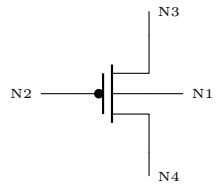
1. `MN n12` source line 25



D N4 = `out` **G** N1 = `vdd` **S** N3 = `in` **B** N2 = `vss`

`w=WLOGICN l=LLOGIC`

2. `MP p12` source line 26



D N4 = `out` **G** N2 = `vss` **S** N3 = `in` **B** N1 = `vdd`

`w=WLOGICP l=LLOGIC`

two_stage_ring

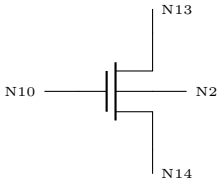
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

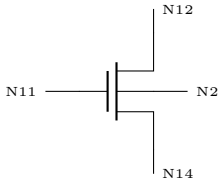
1. `MMAIN1` `n12` source line 86



D N13 = `out_n` **G** N10 = `in_p` **S** N14 = `cs1` **B** N2 = `vss`

`w=WMAIN l=LCORE`

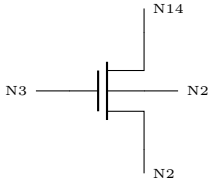
2. `MMAIN2` `n12` source line 87



D N12 = `out_p` **G** N11 = `in_n` **S** N14 = `cs1` **B** N2 = `vss`

`w=WMAIN l=LCORE`

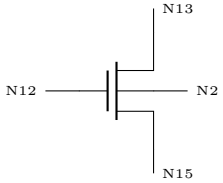
3. `MTAILMAIN` `n12` source line 88



D N14 = `cs1` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`w=WMAIN_TAIL l=LTAIL`

4. `MREGEN1` `n12` source line 91



D N13 = `out_n` **G** N12 = `out_p` **S** N15 = `cs2` **B** N2 = `vss`

`w=WREGEN l=LCORE`

two_stage_ring

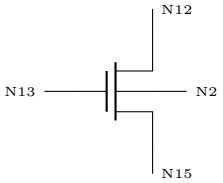
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

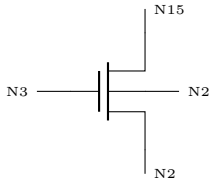
5. `MREGEN2` `n12` source line 92



D N12 = `out_p` **G** N13 = `out_n` **S** N15 = `cs2` **B** N2 = `vss`

`w=WREGEN l=LCORE`

6. `MTAILREGEN` `n12` source line 93



D N15 = `cs2` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`w=WREGEN_TAIL l=LTAIL`

7. `RNEG` `resistor` source line 95



1 N13 = `out_n` **2** N1 = `vdd`

`r=RLOAD`

8. `RPOS` `resistor` source line 96



1 N12 = `out_p` **2** N1 = `vdd`

`r=RLOAD`

two_stage_ring

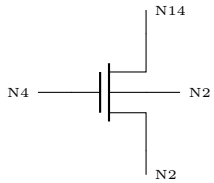
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

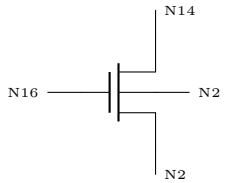
9. `MFINE n12` source line 99



D N14 = `cs1` **G** N4 = `ctrl` **S** N2 = `vss` **B** N2 = `vss`

`w=WFINE l=LFINE`

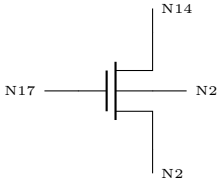
10. `MSUP0 n12` source line 100



D N14 = `cs1` **G** N16 = `g8up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=1`

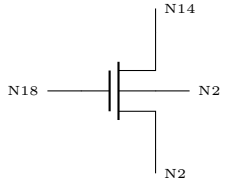
11. `MSUP1 n12` source line 101



D N14 = `cs1` **G** N17 = `g16up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=2`

12. `MSUP2 n12` source line 102



D N14 = `cs1` **G** N18 = `g24up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=3`

two_stage_ring

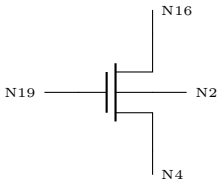
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

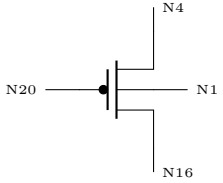
13. `MSUPCTRL0 n12` source line 104



D N16 = `g8up` **G** N19 = `s8up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

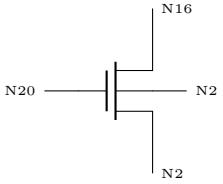
14. `PSUPCTRL0 p12` source line 105



D N16 = `g8up` **G** N20 = `s8up_n` **S** N4 = `ctrl` **B** N1 = `vdd`

`w=1.5u l=LCORE`

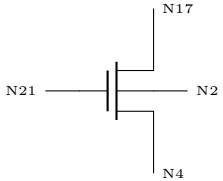
15. `MSUPGND0 n12` source line 106



D N16 = `g8up` **G** N20 = `s8up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

16. `MSUPCTRL1 n12` source line 107



D N17 = `g16up` **G** N21 = `s16up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

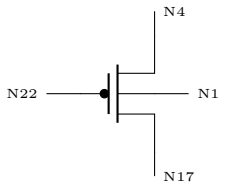
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

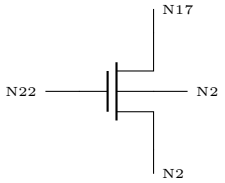
17. `PSUPCTRL1` `p12` source line 108



D N17 = `g16up` **G** N22 = `s16up_n` **S** N4 = `ctrl` **B** N1 = `vdd`

`w=1.5u l=LCORE`

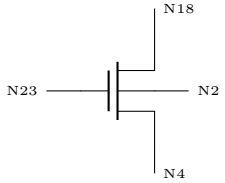
18. `MSUPGND1` `n12` source line 109



D N17 = `g16up` **G** N22 = `s16up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

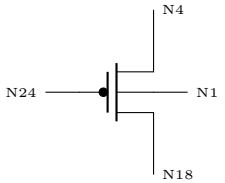
19. `MSUPCTRL2` `n12` source line 110



D N18 = `g24up` **G** N23 = `s24up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

20. `PSUPCTRL2` `p12` source line 111



D N18 = `g24up` **G** N24 = `s24up_n` **S** N4 = `ctrl` **B** N1 = `vdd`

`w=1.5u l=LCORE`

two_stage_ring

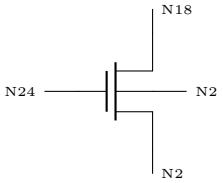
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

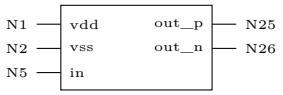
21. `MSUPGND2 n12` source line 112



D N18 = `g24up` **G** N24 = `s24up_n` **S** N2 = `vss` **B** N2 = `vss`

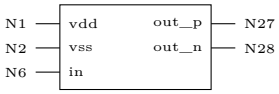
`w=WSW l=LCORE`

22. `XSTD0 s_to_d` source line 114



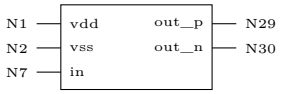
vdd N1 = `vdd` **vss** N2 = `vss` **in** N5 = `s0` **out_p** N25 = `s0_p` **out_n** N26 = `s0_n`

23. `XSTD1 s_to_d` source line 115



vdd N1 = `vdd` **vss** N2 = `vss` **in** N6 = `s1` **out_p** N27 = `s1_p` **out_n** N28 = `s1_n`

24. `XSTD2 s_to_d` source line 116



vdd N1 = `vdd` **vss** N2 = `vss` **in** N7 = `s2` **out_p** N29 = `s2_p` **out_n** N30 = `s2_n`

two_stage_ring

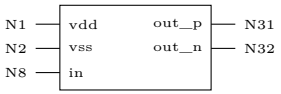
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

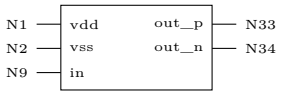
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

25. `XSTD3` `s_to_d` source line 117



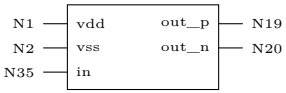
vdd N1 = `vdd` **vss** N2 = `vss` **in** N8 = `s3` **out_p** N31 = `s3_p` **out_n** N32 = `s3_n`

26. `XSTD4` `s_to_d` source line 118



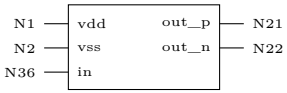
vdd N1 = `vdd` **vss** N2 = `vss` **in** N9 = `s4` **out_p** N33 = `s4_p` **out_n** N34 = `s4_n`

27. `XSTD8` `s_to_d` source line 119



vdd N1 = `vdd` **vss** N2 = `vss` **in** N35 = `s8up` **out_p** N19 = `s8up_p`
out_n N20 = `s8up_n`

28. `XSTD16` `s_to_d` source line 120



vdd N1 = `vdd` **vss** N2 = `vss` **in** N36 = `s16up` **out_p** N21 = `s16up_p`
out_n N22 = `s16up_n`

two_stage_ring

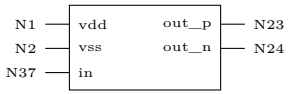
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

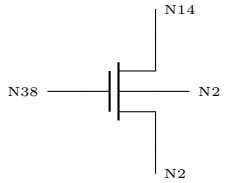
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

29. `XSTD24 s_to_d` source line 121



vdd N1 = `vdd` **vss** N2 = `vss` **in** N37 = `s24up` **out_p** N23 = `s24up_p`
out_n N24 = `s24up_n`

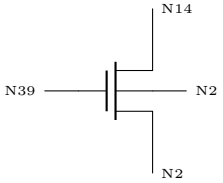
30. `MBAND0 n12` source line 123



D N14 = `cs1` **G** N38 = `g0` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=1`

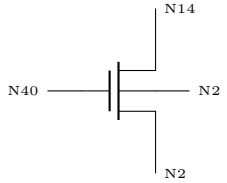
31. `MBAND1 n12` source line 124



D N14 = `cs1` **G** N39 = `g1` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=2`

32. `MBAND2 n12` source line 125



D N14 = `cs1` **G** N40 = `g2` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=4`

two_stage_ring

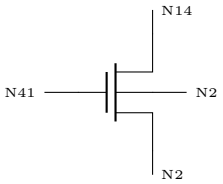
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

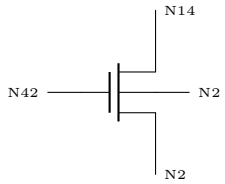
33. `MBAND3 n12` source line 126



D N14 = `cs1` **G** N41 = `g3` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=8`

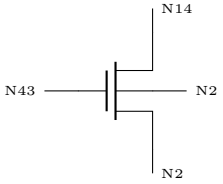
34. `MBAND4 n12` source line 127



D N14 = `cs1` **G** N42 = `g4` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=16`

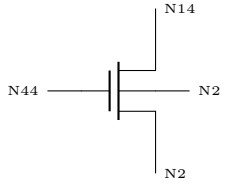
35. `MBAND24UP n12` source line 130



D N14 = `cs1` **G** N43 = `g5` **S** N2 = `vss` **B** N2 = `vss`

`w=WEXTRA24 l=LTAIL`

36. `MBAND16UP n12` source line 131



D N14 = `cs1` **G** N44 = `g6` **S** N2 = `vss` **B** N2 = `vss`

`w=WEXTRA16 l=LTAIL`

two_stage_ring

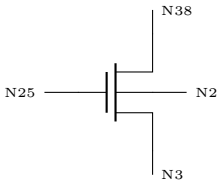
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

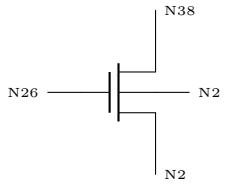
37. `MSWBN0` `n12` source line 133



D N38 = `g0` **G** N25 = `s0_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

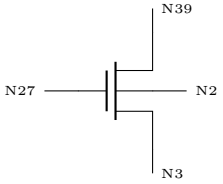
38. `MSWGND0` `n12` source line 134



D N38 = `g0` **G** N26 = `s0_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

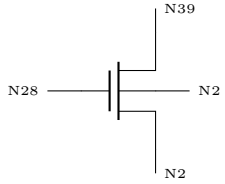
39. `MSWBN1` `n12` source line 135



D N39 = `g1` **G** N27 = `s1_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

40. `MSWGND1` `n12` source line 136



D N39 = `g1` **G** N28 = `s1_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

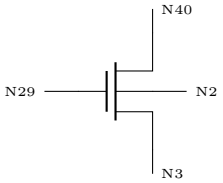
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

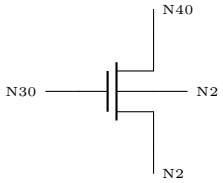
41. `MSWBN2 n12` source line 137



D N40 = `g2` **G** N29 = `s2_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

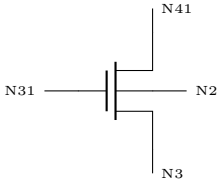
42. `MSWGND2 n12` source line 138



D N40 = `g2` **G** N30 = `s2_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

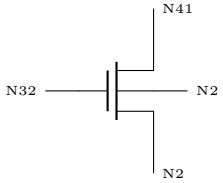
43. `MSWBN3 n12` source line 139



D N41 = `g3` **G** N31 = `s3_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

44. `MSWGND3 n12` source line 140



D N41 = `g3` **G** N32 = `s3_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

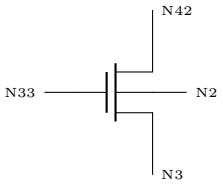
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

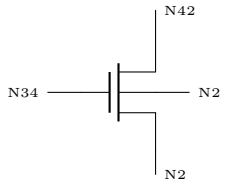
45. `MSWBN4` `n12` source line 141



D N42 = `g4` **G** N33 = `s4_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

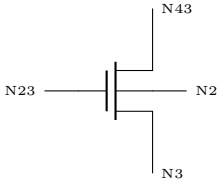
46. `MSWGND4` `n12` source line 142



D N42 = `g4` **G** N34 = `s4_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

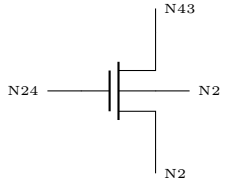
47. `MSWBN5` `n12` source line 143



D N43 = `g5` **G** N23 = `s24up_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

48. `MSWGND5` `n12` source line 144



D N43 = `g5` **G** N24 = `s24up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

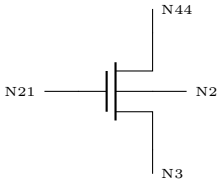
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

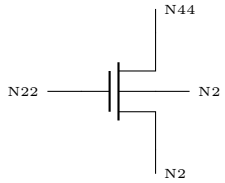
49. `MSWBN6` `n12` source line 145



D N44 = `g6` **G** N21 = `s16up_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

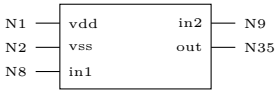
50. `MSWGND6` `n12` source line 146



D N44 = `g6` **G** N22 = `s16up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

51. `XOR8` `or2` source line 149



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N8 = `s3` **in2** N9 = `s4` **out** N35 = `s8up`

52. `XBUF16` `bf_2inv` source line 150



vdd N1 = `vdd` **vss** N2 = `vss` **in** N9 = `s4` **out** N36 = `s16up`

two_stage_ring

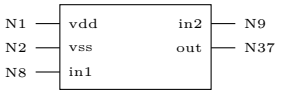
Family `two_stage_ring-66962a1406d1` Declaration lines 84–152

Source `project/netlists/phase2_l3_b11_c0_v1000.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

53. `XAND24` `and2` source line 151



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N8 = `s3` **in2** N9 = `s4` **out** N37 = `s24up`

two_stage_ring

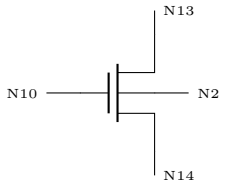
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

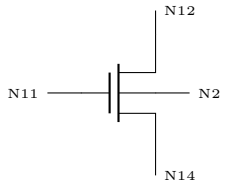
1. `XMAIN1` `n12_ckt_rf` source line 65



D N13 = `out_n` **G** N10 = `in_p` **S** N14 = `cs1` **B** N2 = `vss`

`lr=LCORE wr=3u nf=12 sar=0.4u sbr=0.4u mr=1 mismod=0`

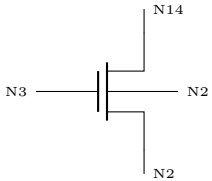
2. `XMAIN2` `n12_ckt_rf` source line 66



D N12 = `out_p` **G** N11 = `in_n` **S** N14 = `cs1` **B** N2 = `vss`

`lr=LCORE wr=3u nf=12 sar=0.4u sbr=0.4u mr=1 mismod=0`

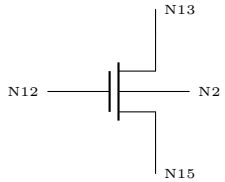
3. `XTAILMAIN` `n12_ckt_rf` source line 67



D N14 = `cs1` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`lr=LTAIL wr=3u nf=12 sar=0.4u sbr=0.4u mr=1 mismod=0`

4. `XREGEN1` `n12_ckt_rf` source line 68



D N13 = `out_n` **G** N12 = `out_p` **S** N15 = `cs2` **B** N2 = `vss`

`lr=LCORE wr=3u nf=8 sar=0.4u sbr=0.4u mr=1 mismod=0`

two_stage_ring

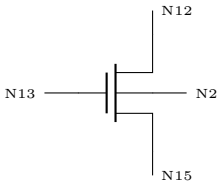
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

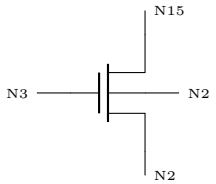
5. `XREGEN2` `n12_ckt_rf` source line 69



D N12 = `out_p` **G** N13 = `out_n` **S** N15 = `cs2` **B** N2 = `vss`

`lr=LCORE wr=3u nf=8 sar=0.4u sbr=0.4u mr=1 mismod=0`

6. `XTAILREGEN` `n12_ckt_rf` source line 70



D N15 = `cs2` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`lr=LTAIL wr=3u nf=16 sar=0.4u sbr=0.4u mr=1 mismod=0`

7. `RNEG` `resistor` source line 71



1 N13 = `out_n` **2** N1 = `vdd`

`r=RLOAD`

8. `RPOS` `resistor` source line 72



1 N12 = `out_p` **2** N1 = `vdd`

`r=RLOAD`

two_stage_ring

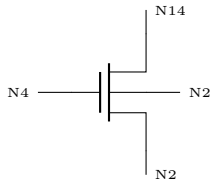
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

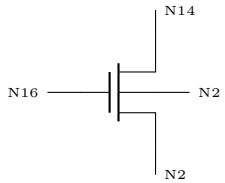
9. `MFINE n12` source line 73



D N14 = `cs1` **G** N4 = `ctrl` **S** N2 = `vss` **B** N2 = `vss`

`w=WFINE l=LFINE`

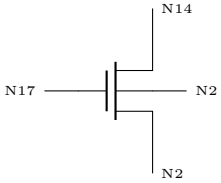
10. `MSUP0 n12` source line 74



D N14 = `cs1` **G** N16 = `g8up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=1`

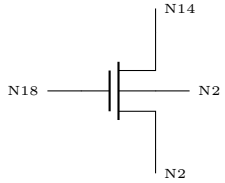
11. `MSUP1 n12` source line 75



D N14 = `cs1` **G** N17 = `g16up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=2`

12. `MSUP2 n12` source line 76



D N14 = `cs1` **G** N18 = `g24up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=3`

two_stage_ring

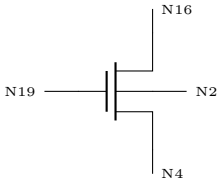
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

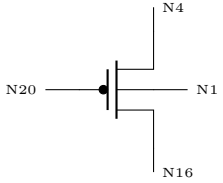
13. `MSUPCTRL0 n12` source line 77



D N16 = `g8up` **G** N19 = `s8up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

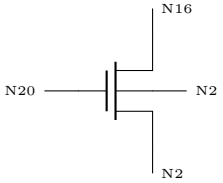
14. `PSUPCTRL0 p12` source line 78



D N16 = `g8up` **G** N20 = `s8up_n` **S** N4 = `ctrl` **B** N1 = `vdd`

`w=1.5u l=LCORE`

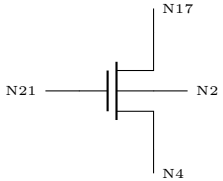
15. `MSUPGND0 n12` source line 79



D N16 = `g8up` **G** N20 = `s8up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

16. `MSUPCTRL1 n12` source line 80



D N17 = `g16up` **G** N21 = `s16up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

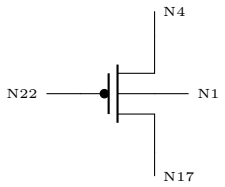
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

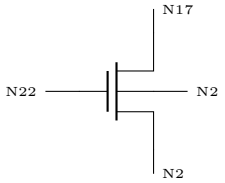
17. `PSUPCTRL1` `p12` source line 81



D N17 = `g16up` **G** N22 = `s16up_n` **S** N4 = `ctrl` **B** N1 = `vdd`

`w=1.5u l=LCORE`

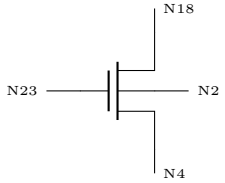
18. `MSUPGND1` `n12` source line 82



D N17 = `g16up` **G** N22 = `s16up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

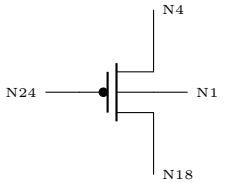
19. `MSUPCTRL2` `n12` source line 83



D N18 = `g24up` **G** N23 = `s24up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

20. `PSUPCTRL2` `p12` source line 84



D N18 = `g24up` **G** N24 = `s24up_n` **S** N4 = `ctrl` **B** N1 = `vdd`

`w=1.5u l=LCORE`

two_stage_ring

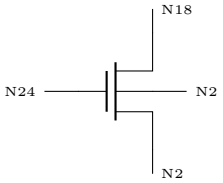
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

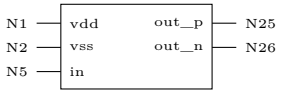
21. `MSUPGND2 n12` source line 85



D N18 = `g24up` **G** N24 = `s24up_n` **S** N2 = `vss` **B** N2 = `vss`

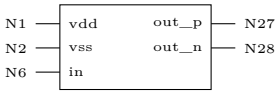
`w=WSW l=LCORE`

22. `XSTD0 s_to_d` source line 86



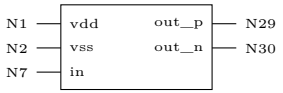
vdd N1 = `vdd` **vss** N2 = `vss` **in** N5 = `s0` **out_p** N25 = `s0_p` **out_n** N26 = `s0_n`

23. `XSTD1 s_to_d` source line 87



vdd N1 = `vdd` **vss** N2 = `vss` **in** N6 = `s1` **out_p** N27 = `s1_p` **out_n** N28 = `s1_n`

24. `XSTD2 s_to_d` source line 88



vdd N1 = `vdd` **vss** N2 = `vss` **in** N7 = `s2` **out_p** N29 = `s2_p` **out_n** N30 = `s2_n`

two_stage_ring

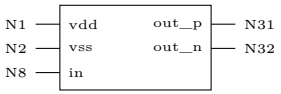
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

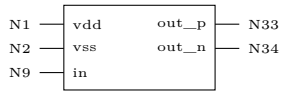
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

25. `XSTD3` `s_to_d` source line 89



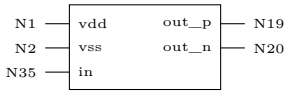
vdd N1 = `vdd` **vss** N2 = `vss` **in** N8 = `s3` **out_p** N31 = `s3_p` **out_n** N32 = `s3_n`

26. `XSTD4` `s_to_d` source line 90



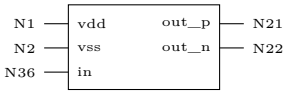
vdd N1 = `vdd` **vss** N2 = `vss` **in** N9 = `s4` **out_p** N33 = `s4_p` **out_n** N34 = `s4_n`

27. `XSTD8` `s_to_d` source line 91



vdd N1 = `vdd` **vss** N2 = `vss` **in** N35 = `s8up` **out_p** N19 = `s8up_p`
out_n N20 = `s8up_n`

28. `XSTD16` `s_to_d` source line 92



vdd N1 = `vdd` **vss** N2 = `vss` **in** N36 = `s16up` **out_p** N21 = `s16up_p`
out_n N22 = `s16up_n`

two_stage_ring

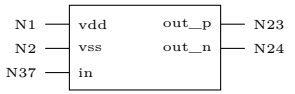
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

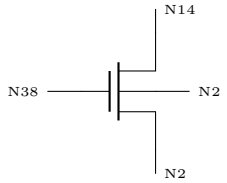
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

29. `XSTD24 s_to_d` source line 93



vdd N1 = `vdd` **vss** N2 = `vss` **in** N37 = `s24up` **out__p** N23 = `s24up_p`
out__n N24 = `s24up_n`

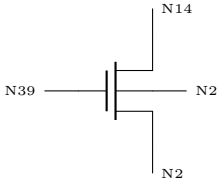
30. `MBAND0 n12` source line 94



D N14 = `cs1` **G** N38 = `g0` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=1`

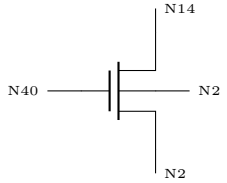
31. `MBAND1 n12` source line 95



D N14 = `cs1` **G** N39 = `g1` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=2`

32. `MBAND2 n12` source line 96



D N14 = `cs1` **G** N40 = `g2` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=4`

two_stage_ring

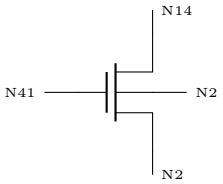
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

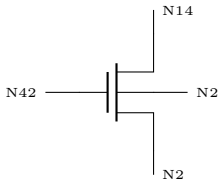
33. `MBAND3 n12` source line 97



D N14 = `cs1` **G** N41 = `g3` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=8`

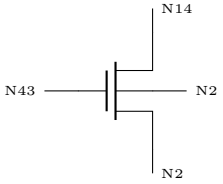
34. `MBAND4 n12` source line 98



D N14 = `cs1` **G** N42 = `g4` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=16`

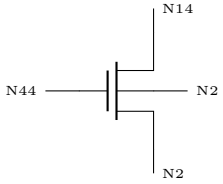
35. `MBAND24UP n12` source line 99



D N14 = `cs1` **G** N43 = `g5` **S** N2 = `vss` **B** N2 = `vss`

`w=WEXTRA24 l=LTAIL`

36. `MBAND16UP n12` source line 100



D N14 = `cs1` **G** N44 = `g6` **S** N2 = `vss` **B** N2 = `vss`

`w=WEXTRA16 l=LTAIL`

two_stage_ring

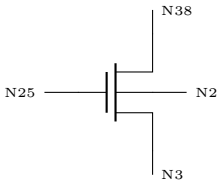
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

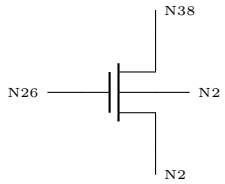
37. `MSWBN0 n12` source line 101



D N38 = `g0` **G** N25 = `s0_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

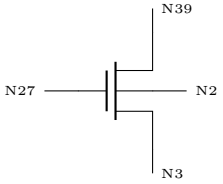
38. `MSWGND0 n12` source line 102



D N38 = `g0` **G** N26 = `s0_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

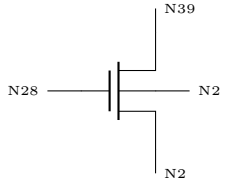
39. `MSWBN1 n12` source line 103



D N39 = `g1` **G** N27 = `s1_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

40. `MSWGND1 n12` source line 104



D N39 = `g1` **G** N28 = `s1_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

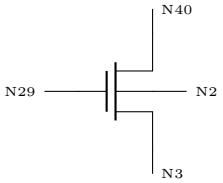
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

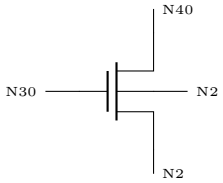
41. `MSWBN2` `n12` source line 105



D N40 = `g2` **G** N29 = `s2_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

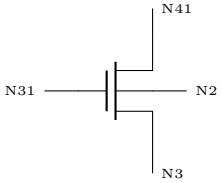
42. `MSWGND2` `n12` source line 106



D N40 = `g2` **G** N30 = `s2_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

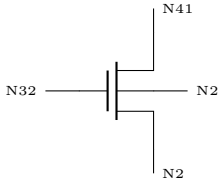
43. `MSWBN3` `n12` source line 107



D N41 = `g3` **G** N31 = `s3_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

44. `MSWGND3` `n12` source line 108



D N41 = `g3` **G** N32 = `s3_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

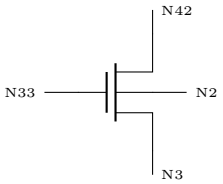
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

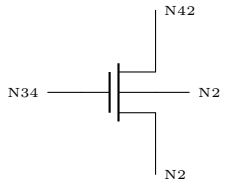
45. `MSWBN4` `n12` source line 109



D N42 = `g4` **G** N33 = `s4_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

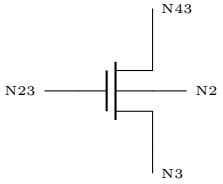
46. `MSWGND4` `n12` source line 110



D N42 = `g4` **G** N34 = `s4_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

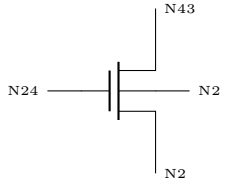
47. `MSWBN5` `n12` source line 111



D N43 = `g5` **G** N23 = `s24up_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

48. `MSWGND5` `n12` source line 112



D N43 = `g5` **G** N24 = `s24up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

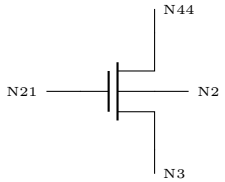
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

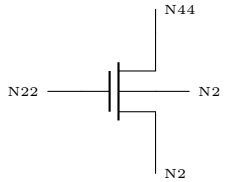
49. `MSWBN6` `n12` source line 113



D N44 = `g6` **G** N21 = `s16up_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

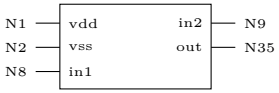
50. `MSWGND6` `n12` source line 114



D N44 = `g6` **G** N22 = `s16up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

51. `XOR8` `or2` source line 115



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N8 = `s3` **in2** N9 = `s4` **out** N35 = `s8up`

52. `XBUF16` `bf_2inv` source line 116



vdd N1 = `vdd` **vss** N2 = `vss` **in** N9 = `s4` **out** N36 = `s16up`

two_stage_ring

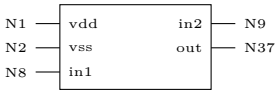
Family `two_stage_ring-6808c58ed797` Declaration lines 64–118

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

53. `XAND24` `and2` source line 117



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N8 = `s3` **in2** N9 = `s4` **out** N37 = `s24up`

two_stage_ring

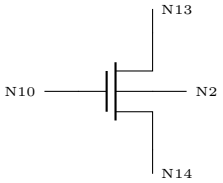
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

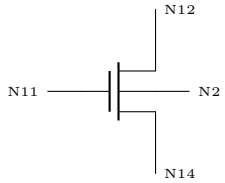
1. `MMAIN1 n12` source line 86



D N13 = `out_n` **G** N10 = `in_p` **S** N14 = `cs1` **B** N2 = `vss`

`w=WMAIN l=LCORE`

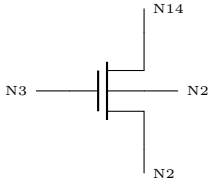
2. `MMAIN2 n12` source line 87



D N12 = `out_p` **G** N11 = `in_n` **S** N14 = `cs1` **B** N2 = `vss`

`w=WMAIN l=LCORE`

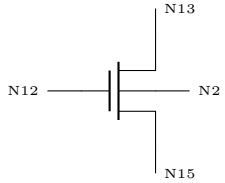
3. `MTAILMAIN n12` source line 88



D N14 = `cs1` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`w=WMAIN_TAIL l=LTAIL`

4. `MREGEN1 n12` source line 91



D N13 = `out_n` **G** N12 = `out_p` **S** N15 = `cs2` **B** N2 = `vss`

`w=WREGEN l=LCORE`

two_stage_ring

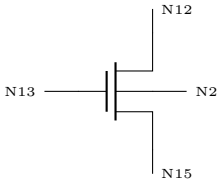
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

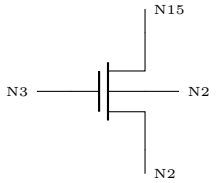
5. `MREGEN2` `n12` source line 92



D N12 = `out_p` **G** N13 = `out_n` **S** N15 = `cs2` **B** N2 = `vss`

`w=WREGEN l=LCORE`

6. `MTAILREGEN` `n12` source line 93



D N15 = `cs2` **G** N3 = `bn` **S** N2 = `vss` **B** N2 = `vss`

`w=WREGEN_TAIL l=LTAIL`

7. `RNEG` `resistor` source line 95



1 N13 = `out_n` **2** N1 = `vdd`

`r=RLOAD`

8. `RPOS` `resistor` source line 96



1 N12 = `out_p` **2** N1 = `vdd`

`r=RLOAD`

two_stage_ring

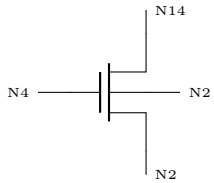
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

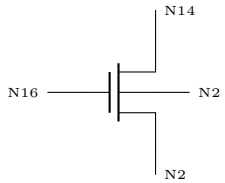
9. `MFINE n12` source line 99



D N14 = `cs1` **G** N4 = `ctrl` **S** N2 = `vss` **B** N2 = `vss`

`w=WFINE l=LFINE`

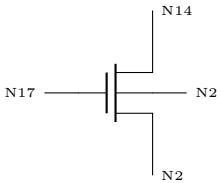
10. `MSUP0 n12` source line 100



D N14 = `cs1` **G** N16 = `g8up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=1`

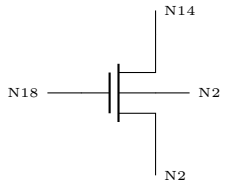
11. `MSUP1 n12` source line 101



D N14 = `cs1` **G** N17 = `g16up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=2`

12. `MSUP2 n12` source line 102



D N14 = `cs1` **G** N18 = `g24up` **S** N2 = `vss` **B** N2 = `vss`

`w=WSUP l=LFINE m=3`

two_stage_ring

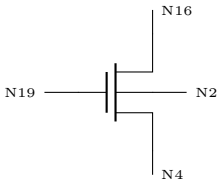
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

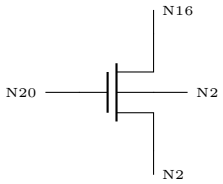
13. `MSUPCTRL0 n12` source line 104



D N16 = `g8up` **G** N19 = `s8up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

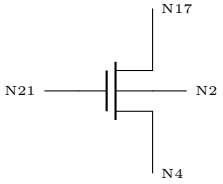
14. `MSUPGND0 n12` source line 105



D N16 = `g8up` **G** N20 = `s8up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

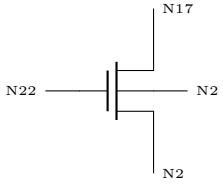
15. `MSUPCTRL1 n12` source line 106



D N17 = `g16up` **G** N21 = `s16up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

16. `MSUPGND1 n12` source line 107



D N17 = `g16up` **G** N22 = `s16up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

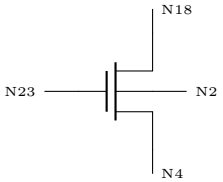
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

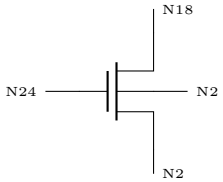
17. `MSUPCTRL2 n12` source line 108



D N18 = `g24up` **G** N23 = `s24up_p` **S** N4 = `ctrl` **B** N2 = `vss`

`w=WSW l=LCORE`

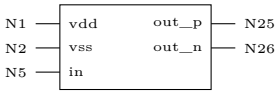
18. `MSUPGND2 n12` source line 109



D N18 = `g24up` **G** N24 = `s24up_n` **S** N2 = `vss` **B** N2 = `vss`

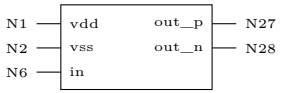
`w=WSW l=LCORE`

19. `XSTD0 s_to_d` source line 111



vdd N1 = `vdd` **vss** N2 = `vss` **in** N5 = `s0` **out_p** N25 = `s0_p` **out_n** N26 = `s0_n`

20. `XSTD1 s_to_d` source line 112



vdd N1 = `vdd` **vss** N2 = `vss` **in** N6 = `s1` **out_p** N27 = `s1_p` **out_n** N28 = `s1_n`

two_stage_ring

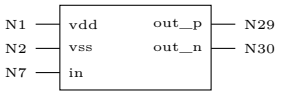
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

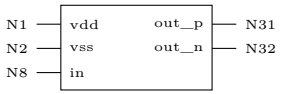
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

21. `XSTD2` `s_to_d` source line 113



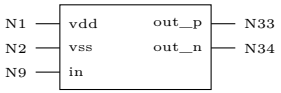
vdd N1 = `vdd` **vss** N2 = `vss` **in** N7 = `s2` **out_p** N29 = `s2_p` **out_n** N30 = `s2_n`

22. `XSTD3` `s_to_d` source line 114



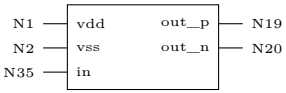
vdd N1 = `vdd` **vss** N2 = `vss` **in** N8 = `s3` **out_p** N31 = `s3_p` **out_n** N32 = `s3_n`

23. `XSTD4` `s_to_d` source line 115



vdd N1 = `vdd` **vss** N2 = `vss` **in** N9 = `s4` **out_p** N33 = `s4_p` **out_n** N34 = `s4_n`

24. `XSTD8` `s_to_d` source line 116



vdd N1 = `vdd` **vss** N2 = `vss` **in** N35 = `s8up` **out_p** N19 = `s8up_p` **out_n** N20 = `s8up_n`

two_stage_ring

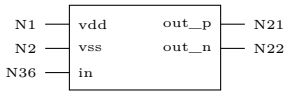
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

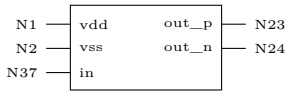
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

25. `XSTD16` `s_to_d` source line 117



vdd N1 = `vdd` **vss** N2 = `vss` **in** N36 = `s16up` **out_p** N21 = `s16up_p`
out_n N22 = `s16up_n`

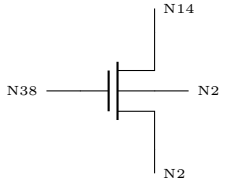
26. `XSTD24` `s_to_d` source line 118



vdd N1 = `vdd` **vss** N2 = `vss` **in** N37 = `s24up` **out_p** N23 = `s24up_p`
out_n N24 = `s24up_n`

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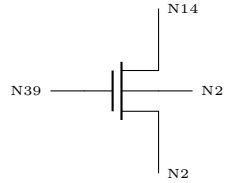
27. `MBANDO` `n12` source line 120



D N14 = `cs1` **G** N38 = `g0` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=1`

28. `MBAND1` `n12` source line 121



D N14 = `cs1` **G** N39 = `g1` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=2`

two_stage_ring

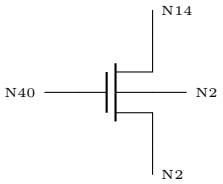
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

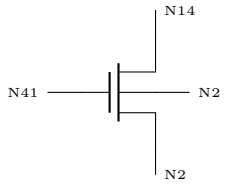
29. `MBAND2 n12` source line 122



D N14 = `cs1` **G** N40 = `g2` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=4`

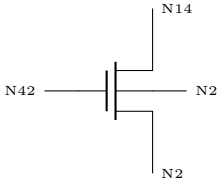
30. `MBAND3 n12` source line 123



D N14 = `cs1` **G** N41 = `g3` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=8`

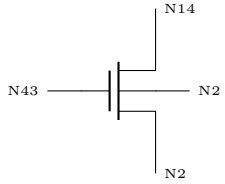
31. `MBAND4 n12` source line 124



D N14 = `cs1` **G** N42 = `g4` **S** N2 = `vss` **B** N2 = `vss`

`w=WBANK l=LTAIL m=16`

32. `MBAND24UP n12` source line 127



D N14 = `cs1` **G** N43 = `g5` **S** N2 = `vss` **B** N2 = `vss`

`w=WEXTRA24 l=LTAIL`

two_stage_ring

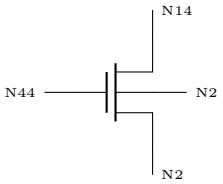
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

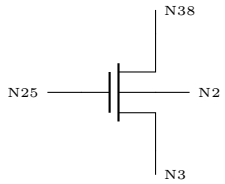
33. `MBAND16UP n12` source line 128



D N14 = `cs1` **G** N44 = `g6` **S** N2 = `vss` **B** N2 = `vss`

`w=WEXTRA16 l=LTAIL`

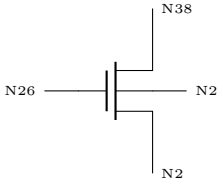
34. `MSWBN0 n12` source line 130



D N38 = `g0` **G** N25 = `s0_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

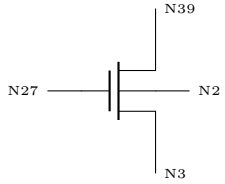
35. `MSWGND0 n12` source line 131



D N38 = `g0` **G** N26 = `s0_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

36. `MSWBN1 n12` source line 132



D N39 = `g1` **G** N27 = `s1_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

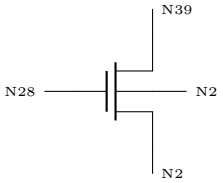
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

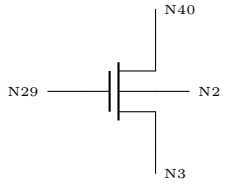
37. `MSWGND1 n12` source line 133



D N39 = `g1` **G** N28 = `s1_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

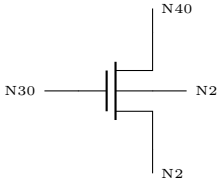
38. `MSWBN2 n12` source line 134



D N40 = `g2` **G** N29 = `s2_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

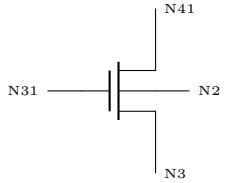
39. `MSWGND2 n12` source line 135



D N40 = `g2` **G** N30 = `s2_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

40. `MSWBN3 n12` source line 136



D N41 = `g3` **G** N31 = `s3_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

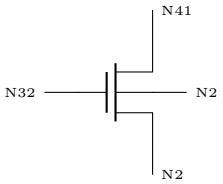
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

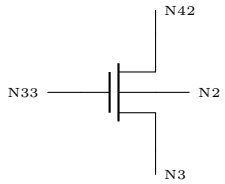
41. `MSWGND3 n12` source line 137



D N41 = `g3` **G** N32 = `s3_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

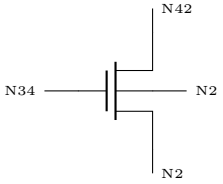
42. `MSWBN4 n12` source line 138



D N42 = `g4` **G** N33 = `s4_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

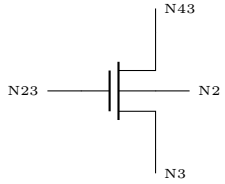
43. `MSWGND4 n12` source line 139



D N42 = `g4` **G** N34 = `s4_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

44. `MSWBN5 n12` source line 140



D N43 = `g5` **G** N23 = `s24up_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

two_stage_ring

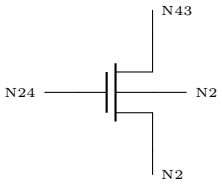
Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

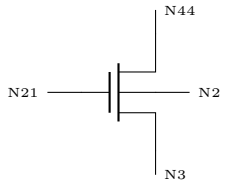
45. `MSWGND5` `n12` source line 141



D N43 = `g5` **G** N24 = `s24up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

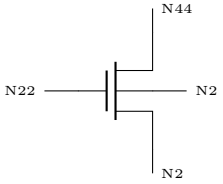
46. `MSWBN6` `n12` source line 142



D N44 = `g6` **G** N21 = `s16up_p` **S** N3 = `bn` **B** N2 = `vss`

`w=WSW l=LCORE`

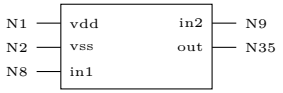
47. `MSWGND6` `n12` source line 143



D N44 = `g6` **G** N22 = `s16up_n` **S** N2 = `vss` **B** N2 = `vss`

`w=WSW l=LCORE`

48. `XOR8` `or2` source line 146



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N8 = `s3` **in2** N9 = `s4` **out** N35 = `s8up`

two_stage_ring

Family `two_stage_ring-ddaed22a25f3` Declaration lines 84–149

Source `project/netlists/vco_bias1p_v6.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `in_p` (N10), `in_n` (N11), `out_p` (N12), `out_n` (N13)

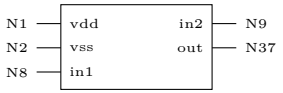
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

49. `XBUF16` `bf_2inv` source line 147



vdd N1 = `vdd` **vss** N2 = `vss` **in** N9 = `s4` **out** N36 = `s16up`

50. `XAND24` `and2` source line 148



vdd N1 = `vdd` **vss** N2 = `vss` **in1** N8 = `s3` **in2** N9 = `s4` **out** N37 = `s24up`

two_stage_ring_vco

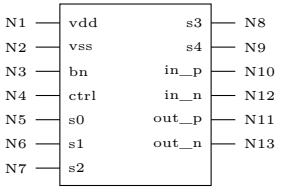
Family `two_stage_ring_vco-6a3dcf32823b` Declaration lines 119–122

Source `project/afc2009_20261007/integration/benches/pll_afc2009_physical_halfvdd_ref10000_4p.scs`

Ports in declaration order `vdd` (N1), `vss` (N2), `bn` (N3), `ctrl` (N4), `s0` (N5), `s1` (N6), `s2` (N7), `s3` (N8), `s4` (N9), `out0` (N10), `out90` (N11), `out180` (N12), `out270` (N13)

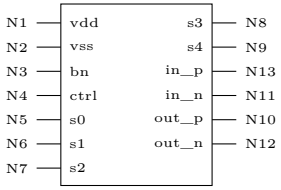
Equal net IDs connect within this module. MOS terminal letters follow the deck order D, G, S, B. Each RF MOS symbol represents the external four-terminal model.

1. `XSTAGE0` `two_stage_ring` source line 120



vdd N1 = `vdd` **vss** N2 = `vss` **bn** N3 = `bn` **ctrl** N4 = `ctrl` **s0** N5 = `s0` **s1** N6 = `s1`
s2 N7 = `s2` **s3** N8 = `s3` **s4** N9 = `s4` **in__p** N10 = `out0` **in__n** N12 = `out180`
out__p N11 = `out90` **out__n** N13 = `out270`

2. `XSTAGE1` `two_stage_ring` source line 121



vdd N1 = `vdd` **vss** N2 = `vss` **bn** N3 = `bn` **ctrl** N4 = `ctrl` **s0** N5 = `s0` **s1** N6 = `s1`
s2 N7 = `s2` **s3** N8 = `s3` **s4** N9 = `s4` **in__p** N13 = `out270` **in__n** N11 = `out90`
out__p N10 = `out0` **out__n** N12 = `out180`